

WavFile_ReadWrite

Jupyter notebook for reading and writing 16-bit .wav files

```
import numpy as np
import IPython.display as ipd
from scipy.io import wavfile

# Read audio signal
Fs, xn16 = wavfile.read('chirp_001.wav')    # read sampling rate Fs and data
xn from .wav file
xn = xn16.astype(np.float32)/2**15          # convert to 32-bit floating point
in range -1.0 to +1.0

# Play xt signal
ipd.Audio(xn, rate=Fs)

<IPython.lib.display.Audio object>

# Modify xn signal
yn = 0.5*xn    # change amplitude

# Write modified yn signal to 16-bit .wav file
yn16 = np.array(2**15*yn, np.int16)
wavfile.write('chirp_001_modified.wav', Fs, yn16)
```